Lakshitha Totakura

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.54
Geetanjali Olympiad School	Class XII (CBSE)	2024	96.80%
Geethanjali Vidyalaya	Class X (CBSE)	2022	99.40%

SCHOLASTIC ACHIEVEMENTS

- **9th IEEE International Test Conference India**- *Fellow Participant*- Participated in technical sessions on emerging research in **IC verification** and testing. Exposure to **scan-chain DFT** architectures and testing in **3D ICs**.
- **IISc Cense Summer School**- *Received* ***Certificate of Distinction***- Learnt **VLSI process flow**, involving bird's eye view of designing, fabrication, packaging and Testing of ICs and micro/nano devices.
- **Selected for Texas Instruments' BYTE Program**, a competitive **analog engineering program**, receiving mentorship from senior TI engineers and hands-on exposure to industry design methodologies.
- **Royal Australian Chemical Institute (RACI):** Awarded **High Distinction** in the **2019 Australian National Chemistry Quiz** (ANCQ) for a **near-perfect score** among **100,000+** participants across **23 countries**.
- **Codeforces** rating (max.) 764

PROJECTS

SIMT GPU Core | GitHub

- Architected a **4-lane SIMT GPU core** with a **custom 6-instruction ISA**, supporting a complete pipeline for instruction fetch, decode, and **parallel SIMD execution.**
- Built a **multi-warp scheduler** utilizing **per-warp banked**, independent **per-thread register files** to manage execution contexts and divergence states.
- Implemented **branch divergence handling** via a **hardware reconvergence stack** to ensure accurate SIMD control flow.
- Integrated a **memory access pattern classifier (coalescing detector)** into the execution pipeline and validated functionality.

5-Stage RV32I Pipelined Processor | GitHub

- Designed and implemented a **5-stage RV32I pipeline** (**IF/ID/EX/MEM/WB**) with a modular datapath and control unit, supporting **R-, I-, S-, and B-type instructions**.
- Implemented **full forwarding (MEM->EX, WB->EX)** and **load-use hazard detection** to resolve data hazards
- Designed a **2-bit branch predictor** with an **8-entry Branch History Table (BHT)** for speculative instruction fetch
- Synthesized and verified pipeline execution through directed RTL simulation using **Icarus Verilog** and **GTKWave**.

Active Thermal Smart Sock | **Indian Air Force sponsored System Design Project**

- Designed a **closed-loop heating system** to mitigate frostbite at **sub zero (-60°C)** and **high altitude** conditions.
- Designed a **custom multi-layer nRF52840 MCU PCB**, integrating battery protection, power regulation, and PWM control.
- Interfaced **TMP117 sensors via I2C** to drive a real-time, dynamic **10W active heating** algorithm.
- Executed **end-to-end development**, initial schematic capture and component selection to layout routing and fabrication.

Inter IIT Tech Meet 14.0 | Drona Aviation

- Selected to represent IITH for the **indoor nano-drone stabilization** problem statement at Inter IIT Tech Meet 14.0.
- Implemented **Kalman filtering (Optical Flow, ToF, IMU)** and **cascaded PID** loops on **Pluto/Magis flight firmware.**

- Validated **flight control stability** on **Pluto X** hardware through extensive real-time flight testing and log analysis.

RF Function Generator | Prof. Naresh Kumar Emani

- Engineered a **custom RF signal generator PCB** with a **JFET Colpitts carrier oscillator** (LC-tuned up to **5 MHz**).
- Built an **830 Hz** BJT RC phase-shift **audio oscillator** and collector stage for **amplitude modulation (AM).**
- Implemented a **TL071 audio buffer** and an **LM6172** high-speed **comparator squarer** for frequency measurement.

SKILLS

- **Design and Tools:** RTL Design | FPGA Design | PCB Design | Xilinx Vivado | Altium Designer | Kicad | SolidEdge | LTSpice | Icarus Verilog | Arduino IDE | GTKWave | Git | PYNQ-Z2 (Zynq-7020)

- **Programming Languages:** Verilog | C | C++ | Python | RISC - V Assembly | LaTeX | MATLAB | SQL | ARM Assembly | ROS2

RELEVANT COURSES

- **Theory**: Computer Architecture | Digital Systems | Analog Electronics | Circuit Network Analysis | Control Theory | Signals and Systems | CMOS processing | SDF | Productivity Tools for IC Design | Numerical Methods for Device Modeling | Machine Learning | DSA | EM | Matrix Theory | Probability and Random Processes

- **Lab:** Electronic System Design | PCB Design | Analog | Digital Systems | Semiconductor Devices | Electrical Circuits

POSITIONS OF RESPONSIBILITY

Domain Head | Elektronica - Electronics and Signal Processing Club, IIT Hyderabad

- Heading the **Embedded Systems Domain** at Elektronica, organizing technical workshops, project-based learning, and mentorship for **30+ student members**.

Core | Robotix (Robotics Club) **|** Embedded Systems domain

- **Krishi Cobot** project for the **e-Yantra Robotics Competition**, employing **ROS 2** for autonomous system integration.

EXTRACURRICULAR

- **Competitive Scrabble player** - Represented IITH in **Inter IIT Cultural Meets 7.0 and 8.0**